

PEDIAIR

Montelukast Granules BP 4mg

Pharmacode

Formulation:

Each Sachet (1g) Contains:
Montelukast Sodium USP 4.16 mg
Eq. to Montelukast 4 mg

Product Description:

White to off-white granular powder.

Pharmacodynamics/ Pharmacokinetics

Pharmacotherapeutic group:

Leukotriene receptor antagonists

ATC-code: R03D C03

Mechanism of action

The cysteinyl leukotrienes (LTC4, LTD4, LTE4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene receptors (CysLT) found in the human airway and cause airway actions, including bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment.

Pharmacodynamic effects

Montelukast is an orally active compound, which binds with high affinity and selectivity to the CysLT1 receptor.

Pharmacokinetics

Absorption: Montelukast is rapidly and nearly completely absorbed following oral administration. The mean oral bioavailability is 64%. The oral bioavailability and Cmax are not influenced by a standard meal.

Distribution: Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8 to 11 liters.

Metabolism: Montelukast is extensively metabolized in the liver by cytochrome P450 isoenzymes CYP3A4 and CYP2C9. Therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6.

Elimination: Montelukast and its metabolites are excreted principally in the faeces via the bile. Elimination Half-life: 2.7 to 5 hours.

Indication:

- For the prophylaxis and chronic treatment of asthma in adults and pediatric patients 12 months of age and older.
- Indicated in adults and pediatric patients 2 years of age and older for the relief of daytime and nighttime symptoms of seasonal allergic rhinitis

Recommended Dosage:

- Montelukast should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.
- Patients with both asthma and allergic rhinitis should take only one sachet daily in the evening.
- Dosage of Pediatric Patients 2 to 5 years of age with Asthma and/or Seasonal Allergic Rhinitis is one sachet of 4 mg oral granules daily.

- Dosage of Pediatric Patients 12 months to 2 years of age with Asthma is one sachet of 4 mg oral granules daily.

Administration of oral granules:

Montelukast oral granules can be administered either directly in the mouth, or mixed with a spoonful of cold or room temperature soft food (e.g., applesauce) or dissolved in 1 teaspoonful (5 mL) of cold or room temperature baby formula or breast milk. The sachet should not be opened until ready to use. After opening the sachet, the full dose of montelukast oral granules must be administered immediately (within 15 minutes). If mixed with food or dissolved in baby formula or breast milk, montelukast oral granules must not be stored for future use. Montelukast oral granules are not intended to be dissolved in any liquid other than baby formula or breast milk for administration. However, liquids may be taken subsequent to administration.

- The therapeutic effect of montelukast on parameters of asthma control occurs within one day. Montelukast oral granules can be taken with or without food. Patients should be advised to continue taking montelukast while their asthma is controlled, as well as during periods of worsening asthma.
- No dosage adjustment is necessary for pediatric patients, for the elderly, for patients with renal insufficiency, or mild-to moderate hepatic impairment, or for patients of either gender.
- Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent.
- Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with Montelukast in Relation to Other Treatments for Asthma:

Montelukast can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:

Bronchodilator Treatments: Montelukast can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with montelukast provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. Montelukast should not be abruptly substituted for inhaled corticosteroids.

Route of Administration: Oral

Contraindications:

Hypersensitivity to the active substance or to any of the excipients.

Warnings and Precautions:

Jauhkan daripada capaian kanak-kanak.

Keep out of reach of children.

Please consult your pharmacist/doctor before taking this product.

The diagnosis of persistent asthma in very young children (12 months – 2 years) should be established by a paediatrician or pulmonologist.

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled beta-agonist should be used. Patients should seek their doctors' advice as soon as possible if they need more inhalations of short-acting beta-agonists than usual.

Montelukast should not be abruptly substituted for inhaled or oral corticosteroids.

In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition, which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.

Neuropsychiatric events (e.g. agitation, aggression, anxiousness, dream abnormalities, hallucinations, depression, disorientation, insomnia, irritability, restlessness, suicidal thinking and behaviour (including suicide), tremor) have occurred. Since other factors may have contributed to these events, it is not known if they are related to Montelukast. Physicians should discuss these adverse experiences with their patients and/or caregivers. Patients and/or caregivers should be instructed to notify their physician if these changes occur. Prescribers should carefully evaluate the risks and benefits of continuing treatment with Montelukast if such events occur.

Interactions with Other Medicaments:

Interactions with Other Medicaments

- Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma and in the treatment of allergic rhinitis.
- Since montelukast is metabolised by CYP 3A4, caution should be exercised, particularly in children, when montelukast is co-administered with enzyme inducers of CYP 3A4, such as phenytoin, phenobarbital and rifampicin.
- Concurrent use of montelukast and phenobarbital may result in decreased bioavailability of montelukast.
- Concurrent use of montelukast and rifampicin may result in decreased bioavailability of montelukast.
- Montelukast is a potent inhibitor of CYP 2C8. Montelukast is not anticipated to markedly alter the metabolism of medicinal products metabolised by this enzyme (e.g., paclitaxel, rosiglitazone, and repaglinide.)
- Concurrent use of gemfibrozil and montelukast may result in elevated montelukast plasma concentrations.
- Concurrent use of prednisone and montelukast may result in severe peripheral edema.
- Co-administration of montelukast with itraconazole, a strong inhibitor of CYP 3A4, resulted in no significant increase in the systemic exposure of montelukast.

Statement on usage during pregnancy and lactation:

Pregnancy: Limited data from available pregnancy databases do not suggest a causal relationship between Montelukast and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience. Montelukast may be used during pregnancy only if it is considered to be clearly essential.

Lactation: It is not known if montelukast is excreted in human milk. Montelukast may be used in breast-feeding mothers only if it is considered to be clearly essential.

Adverse Effects/ Undesirable Effects:

Montelukast oral granules has been generally well tolerated. Side effects, which usually were mild, generally did not require discontinuation of therapy.

Postmarketing Experience

The following side effects have been reported in postmarketing use:

Infections and infestations: upper respiratory infection

Blood and lymphatic system disorders: increased bleeding tendency, thrombocytopenia

Immune system disorders: hypersensitivity reactions including anaphylaxis, very rarely hepatic eosinophilic infiltration

Psychiatric disorders: agitation including aggressive behavior or hostility, anxiousness, depression, disorientation, disturbances in attention, dream abnormalities, hallucinations, insomnia, memory impairment, obsessive-compulsive symptoms, psychomotor hyperactivity (including irritability, restlessness, and tremor), somnambulism, suicidal thinking and behavior (suicidality), tic

Nervous system disorders: dizziness, drowsiness, paraesthesia/hypoesthesia, very rarely seizure

Cardiac disorders: palpitations

Respiratory, thoracic and mediastinal disorders: epistaxis; pulmonary eosinophilia

Gastrointestinal disorders: diarrhea, dyspepsia, nausea, vomiting

Hepatobiliary disorders: increased ALT and AST, very rarely hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury)

Skin and subcutaneous tissue disorders: angioedema, bruising, erythema multiforme, erythema nodosum, pruritus, rash, urticaria

Musculoskeletal and connective tissue disorders: arthralgia, myalgia including muscle cramps

Renal and urinary disorders: enuresis in children

General disorders and administration site conditions: asthenia/fatigue, edema, pyrexia

Effects on Ability to Drive and Use Machine:

Montelukast has no or negligible influence on the ability to drive and use machines. However, individuals have reported drowsiness or dizziness.

Overdose and Treatment:

Symptoms of overdose

The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

Management of overdose

No specific information is available on the treatment of overdose with montelukast. It is not known whether montelukast is dialysable by peritoneal- or haemo-dialysis.

Storage Conditions:

Store at temperature not exceeding 30°C. Protect from light and moisture.

Dosage forms and packaging available:

Granules

28 sachets in a carton

Revision Date: 02/02/2021

XL

Manufactured by :

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Pharmacode